

Test 1 Review

Name(s): _____

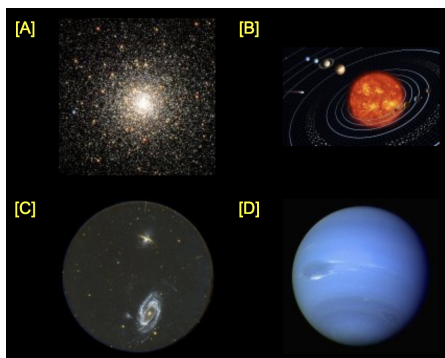
1. If the radius of the Sun is roughly 28 times that of Saturn, the volume of the Sun is roughly _____ that of Saturn.

A. 28 times B. 784 times C. 21,952 times D. Unrelated to

2. The name of our galaxy is _____.

A. Andromeda C. The Large Magellenic Cloud
B. The Milky Way D. The Small Magellenic Cloud

3. Rank the following objects from smallest to largest.



4. If Ceres lies 2.8 A.U. from the Sun, what is the distance between the Sun and Ceres in meters?

5. If Betelgeuse (a star) lies 642.5 light-years from the Sun. How many kilometers separate Betelgeuse and the Sun?

6. Write the number in scientific notation: 0.00234

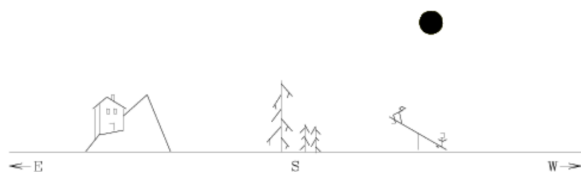
7. Suppose $L = R^2 T^4$. What is the value of T, if $R = 4$ and $L = 256$?

8. The Big Pink skyscraper in Portland has a height of 163 m. Find the order of magnitude of its height.

9. What is the surface area of a soccer ball whose radius is about 0.11 m?
10. The Dwarf planet Eris has a year that is 557 times that of Earth. How long is this in seconds?
11. $10^7 \div 10^3 =$
12. $(10^3)^3 =$
13. $x^4 y^{2.4} = z^{1.3}$ Solve for y.
14. Suppose that Portlandness (P) is related to Coffeeness (C) and Trailblazerness (T) according to $P^3 = 6C^2T^4$. If C doubled, and T became 1.2 times as big, how many times bigger would Portlandness be?
15. The subatomic particles called protons, neutrons, and electrons each carry a positive, neutral, and negative charge respectively. If an atom of standard carbon contains six protons, six neutrons, and seven electrons, what is its net charge?
16. The Radius of Betelgeuse is 1000 times as big as the sun. Its volume is _____times bigger.
17. A 10 kg block is accelerated at 3 m/s^2 . What force acts on the block?
18. If the Moon is located as shown in the figure below, could you observe it in the sky at 2 pm?

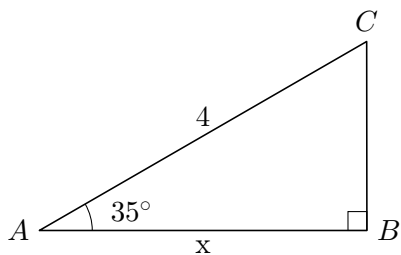


19. If a first quarter Moon rose three hours ago, what time is it?
20. How long, in days, does it take the Moon to shift from a waning gibbous phase to a new phase?
21. This figure shows the location and phase of the Moon as we look from our position in the northern hemisphere toward the southern sky, with the horizon stretching from east (on the left) to west (on the right). If black indicates the un-illuminated portion of the Moon, what time of the night or day is shown?



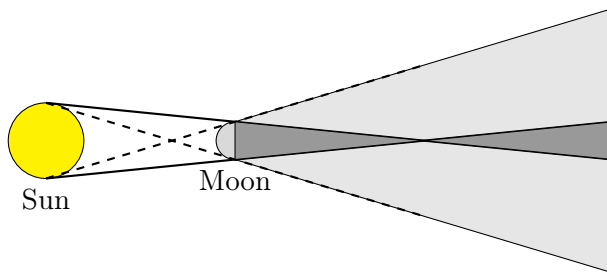
22. If the moon was moved four times further away but the Earth's mass was made twice as big, the Earth's new gravitational force would be _____ times the old force?
23. A car drives down the road and travels 40 km in 2 hours. What is the velocity of the car?
24. What is the Orbital Velocity of Mercury in km/s ? It orbits the sun at a distance of 0.38 AU and has a year equal to 88 Earth days.

25. Solve for x



26. The radius of the Sun is _____ meters, and the distance between the Earth and Sun is _____ meters. If you used a volleyball with a 0.1-m radius to represent the Sun, how far would you need to place it from your model Earth in a scale model of the solar system?

27. Label and name the part of the shadow the Earth would have to be for a Total Solar eclipse to occur.



28. If the Moon is in the waning gibbous phase today, will it be more or less illuminated tomorrow?
29. Why does an ice skater spin faster after they bring in their arms?
30. A total solar eclipse happens approximately every _____ months.